

Tugas Mata Kuliah Pemrograman Website
Laporan Hasil Pengerjaan Aktivitas Mandiri



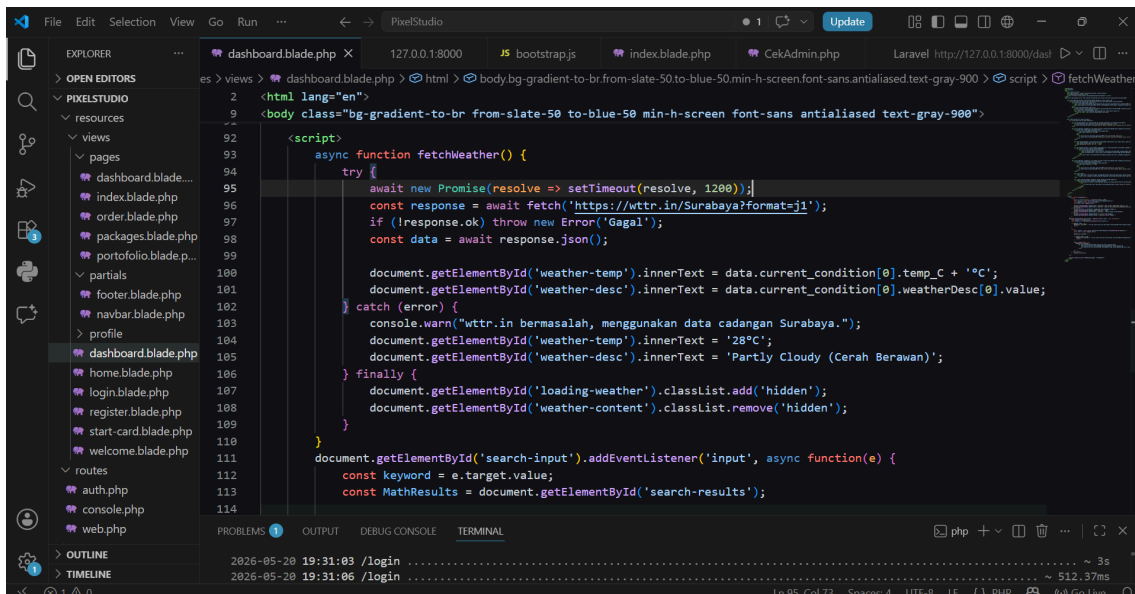
Oleh :

Oktavia Eka Rahmadini (242410101043)

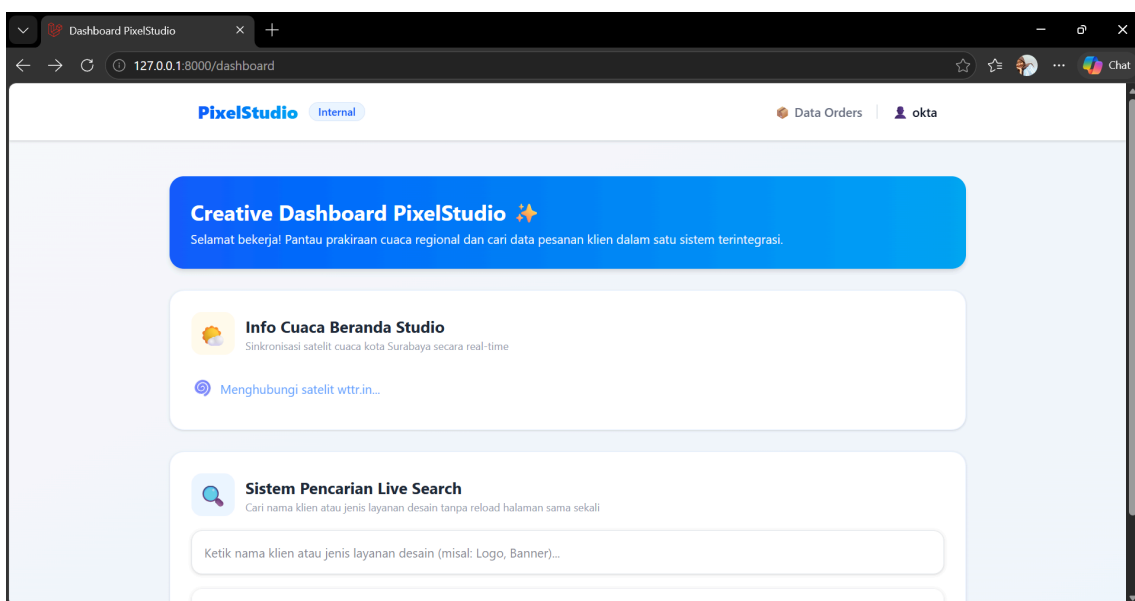
PROGRAM STUDI SISTEM INFORMASI
FAKULTAS ILMU KOMPUTER
UNIVERSITAS JEMBER

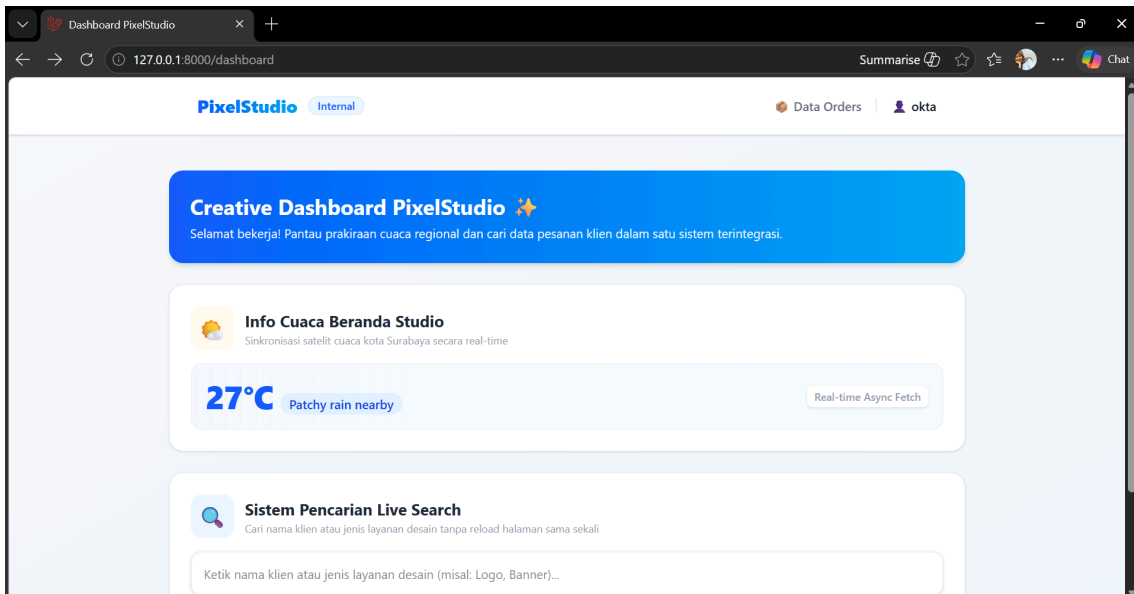
2026

1. Membuat komponen antarmuka (widget) yang mampu menampilkan informasi cuaca secara asinkronus (*Asynchronous*) tanpa mengganggu proses pemuatan elemen halaman utama lainnya.

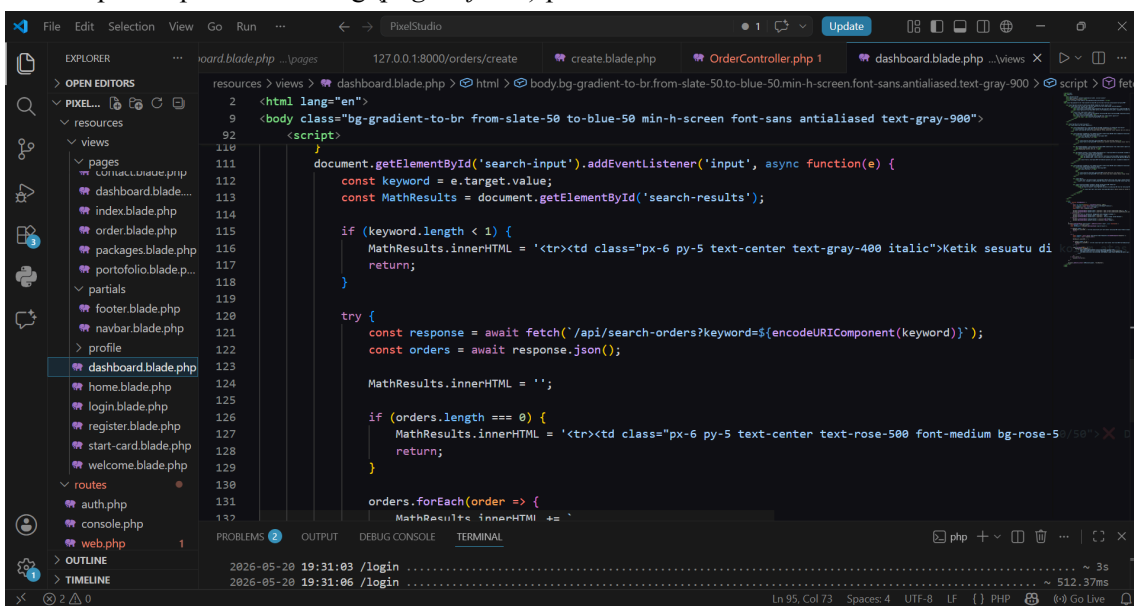


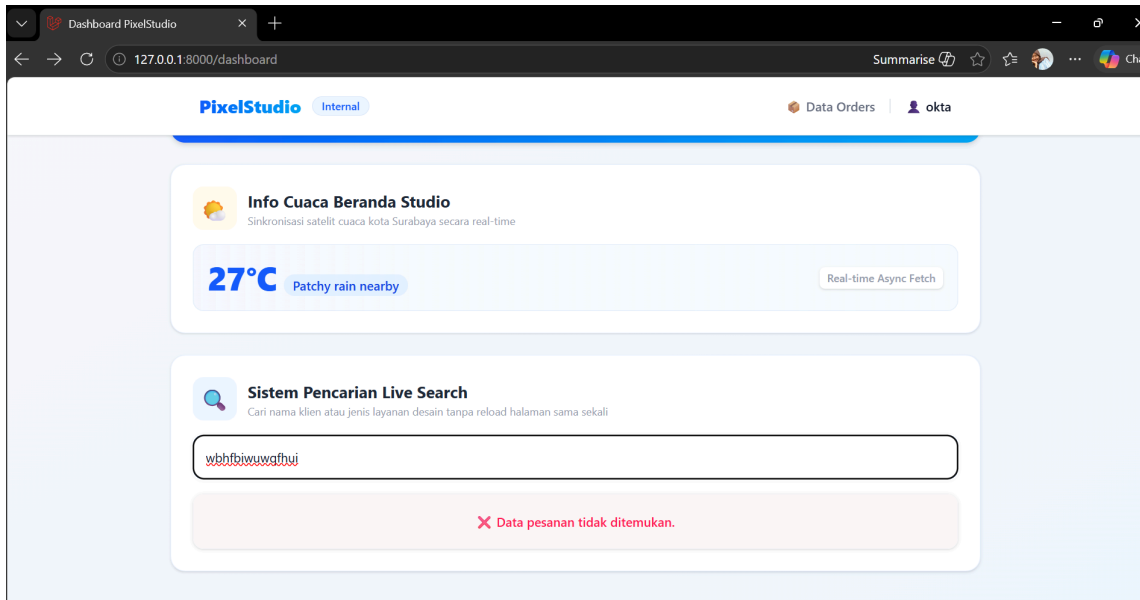
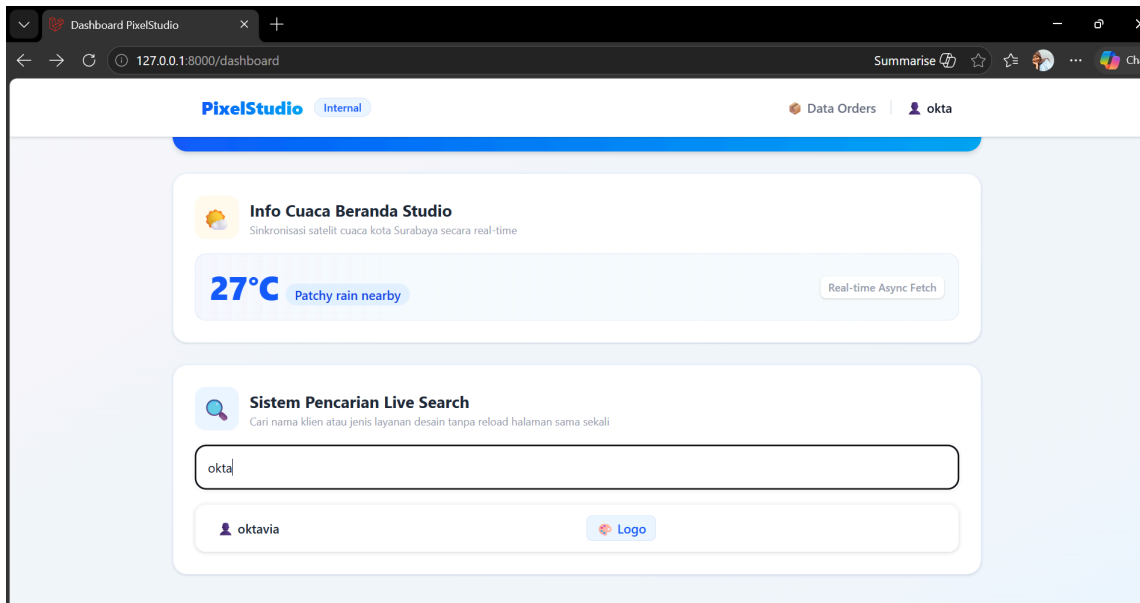
```
es > views > dashboard.blade.php > <html lang="en">
2 <body class="bg-gradient-to-br from-slate-50 to-blue-50 min-h-screen font-sans antialiased text-gray-900">
9
100 <script>
101   async function fetchWeather() {
102     try {
103       await new Promise(resolve => setTimeout(resolve, 1200));
104       const response = await fetch('https://wttr.in/Surabaya?format=json');
105       if (!response.ok) throw new Error('Gagal');
106       const data = await response.json();
107
108       document.getElementById('weather-temp').innerText = data.current_condition[0].temp_C + '°C';
109       document.getElementById('weather-desc').innerText = data.current_condition[0].weatherDesc[0].value;
110     } catch (error) {
111       console.warn("wttr.in bermasalah, menggunakan data cadangan Surabaya.");
112       document.getElementById('weather-temp').innerText = '28°C';
113       document.getElementById('weather-desc').innerText = 'Partly Cloudy (Cerah Berawan)';
114     } finally {
115       document.getElementById('loading-weather').classList.add('hidden');
116       document.getElementById('weather-content').classList.remove('hidden');
117     }
118   }
119
120   document.getElementById('search-input').addEventListener('input', async function(e) {
121     const keyword = e.target.value;
122     const MathResults = document.getElementById('search-results');
123   });
124 }
```



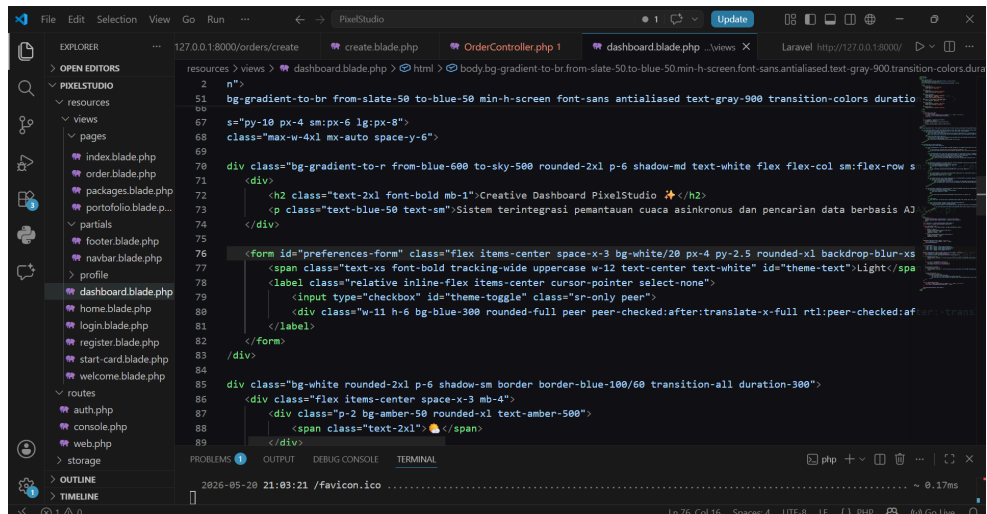


- Menyediakan kolom pencarian data yang dapat menyaring dan menampilkan hasil pencarian dari basis data secara langsung (*live*) ke dalam tabel, tanpa memerlukan proses pemuatan ulang (*page refresh*) pada browser.



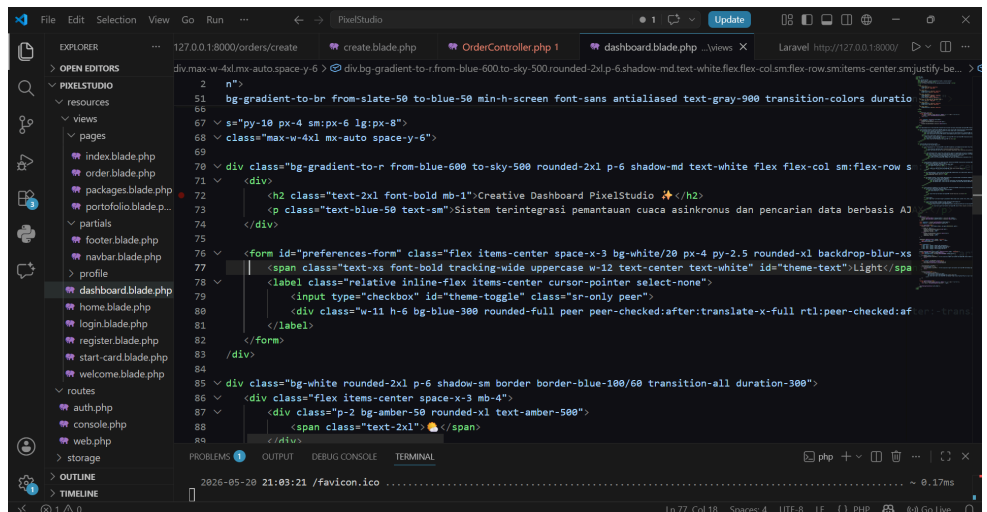


3. Membuat fitur pengubah tema warna halaman (Terang/Gelap) yang konfigurasinya wajib disimpan ke dalam sistem penyimpanan lokal browser (*Cookie*) agar pilihan pengguna tidak hilang saat halaman disegarkan atau ditutup.
 - a. Elemen `<form>` untuk input preferensi



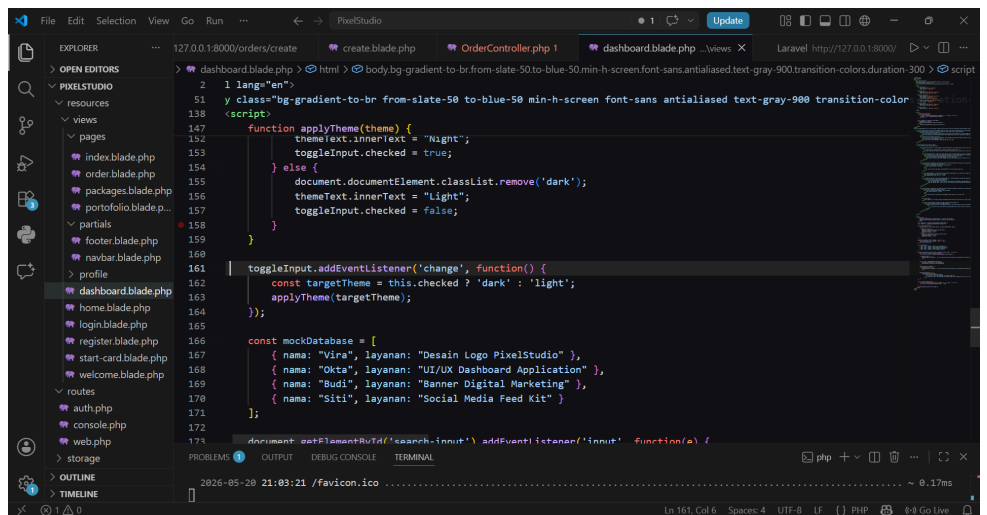
```
2 n">
51 bg-gradient-to-br from-slate-50 to-blue-50 min-h-screen font-sans antialiased text-gray-900 transition-colors duration-300
52
53 s="py-10 px-4 sm:px-6 lg:px-8"
54 class="max-w-4xl mx-auto space-y-6">
55
56 div class="bg-gradient-to-r from-blue-600 to-sky-500 rounded-2xl p-6 shadow-md text-white flex flex-col sm:flex-row space-y-4 sm:space-x-4"
57 <div>
58 <h2 class="text-2xl font-bold mb-1">Creative Dashboard PixelStudio
59 <p class="text-blue-50 text-sm">Sistem terintegrasi pemantauan cuaca asinkronus dan pencarian data berbasis AI
60 </div>
61
62 <form id="preferences-form" class="flex items-center space-x-3 bg-white/20 px-4 py-2.5 rounded-xl backdrop-blur-xs"
63 <span class="text-xs font-bold tracking-wide uppercase w-12 text-center text-white" id="theme-text">Light
64 <label class="relative inline-flex items-center cursor-pointer select-none"
65 <input type="checkbox" id="theme-toggle" class="sr-only peer"
66 </label>
67 </form>
68 </div>
69
70 div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300"
71 <div class="flex items-center space-x-3 mb-4">
72 <div class="p-2 bg-amber-50 rounded-xl text-amber-500">
73 <span class="text-2xl">🔥
74 </span>
75 </div>
76
77 </div>
78
79 </div>
80
81 </div>
82
83 </div>
84
85 </div>
86
87 </div>
88
89 </div>
```

b. Input teks/dropdown untuk nama tema



```
2 n">
51 bg-gradient-to-br from-slate-50 to-blue-50 min-h-screen font-sans antialiased text-gray-900 transition-colors duration-300
52
53 s="py-10 px-4 sm:px-6 lg:px-8"
54 class="max-w-4xl mx-auto space-y-6">
55
56 div class="bg-gradient-to-r from-blue-600 to-sky-500 rounded-2xl p-6 shadow-md text-white flex flex-col sm:flex-row space-y-4 sm:space-x-4"
57 <div>
58 <h2 class="text-2xl font-bold mb-1">Creative Dashboard PixelStudio
59 <p class="text-blue-50 text-sm">Sistem terintegrasi pemantauan cuaca asinkronus dan pencarian data berbasis AI
60 </div>
61
62 <form id="preferences-form" class="flex items-center space-x-3 bg-white/20 px-4 py-2.5 rounded-xl backdrop-blur-xs"
63 <span class="text-xs font-bold tracking-wide uppercase w-12 text-center text-white" id="theme-text">Light
64 <label class="relative inline-flex items-center cursor-pointer select-none"
65 <input type="checkbox" id="theme-toggle" class="sr-only peer"
66 </label>
67 </form>
68 </div>
69
70 div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300"
71 <div class="flex items-center space-x-3 mb-4">
72 <div class="p-2 bg-amber-50 rounded-xl text-amber-500">
73 <span class="text-2xl">🔥
74 </span>
75 </div>
76
77 </div>
78
79 </div>
80
81 </div>
82
83 </div>
84
85 </div>
86
87 </div>
88
89 </div>
```

c. Event Listener (submit/change)



```
2 l lang="en">
51 y class="bg-gradient-to-br from-slate-50 to-blue-50 min-h-screen font-sans antialiased text-gray-900 transition-colors duration-300"
52
53 <script>
54 function applyTheme(theme) {
55 themeText.innerText = "Night";
56 toggleInput.checked = true;
57 } else {
58 document.documentElement.classList.remove('dark');
59 themeText.innerText = "Light";
60 toggleInput.checked = false;
61 }
62
63 toggleInput.addEventListener('change', function() {
64 const targetTheme = this.checked ? 'dark' : 'light';
65 applyTheme(targetTheme);
66 });
67
68 const mockDatabase = [
69 { name: "Vira", layanan: "Desain Logo PixelStudio" },
70 { name: "Dita", layanan: "UI/UX Dashboard Application" },
71 { name: "Budi", layanan: "Banner Digital Marketing" },
72 { name: "Siti", layanan: "Social Media Feed Kit" }
73 ];
74
75 document.getElementById('search-input').addEventListener('input', function(e) {
76
77
78
79
80
81
82
83
84
85
86
87
88
89
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100
```

d. Fungsi JavaScript untuk menyimpan ke cookie

```

1  lang="en">
51  y class="bg-gradient-to-br from-slate-50 to-blue-50 min-h-screen font-sans antialiased text-gray-900 transition-colors duration-300"
138  <script>
139
140  setTimeout(() => {
141    document.getElementById('weather-content').classList.remove('hidden');
142  }, 1200);
143
144  const toggleInput = document.getElementById('theme-toggle');
145  const themeText = document.getElementById('theme-text');
146
147  function applyTheme(theme) {
148    document.cookie = `pref_theme=${theme}; max-age=${60*60*24*30}; path=/`;
149
150    if (theme === 'dark') {
151      document.documentElement.classList.add('dark');
152      themeText.innerText = "Night";
153      toggleInput.checked = true;
154    } else {
155      document.documentElement.classList.remove('dark');
156      themeText.innerText = "Light";
157      toggleInput.checked = false;
158    }
159  }
160
161  toggleInput.addEventListener('change', function() {

```

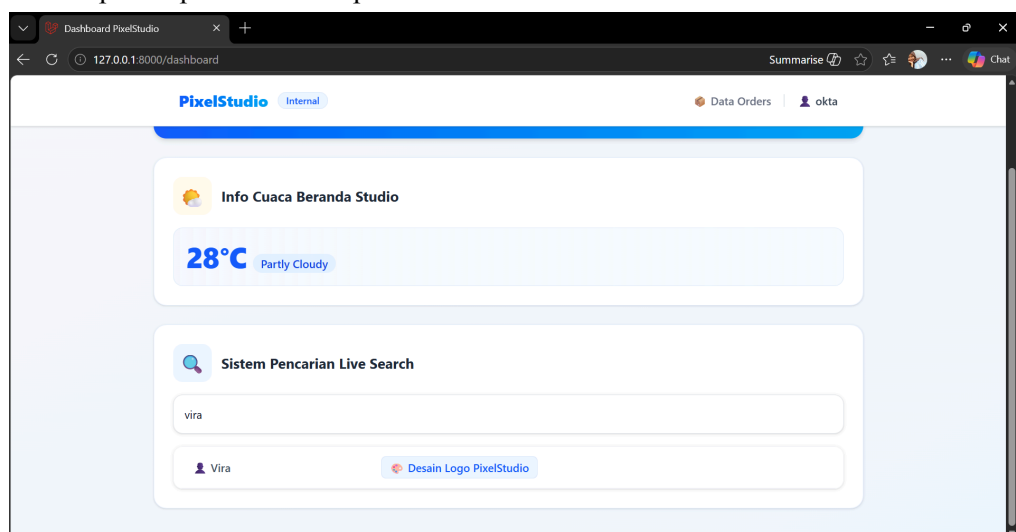
e. Logika percabangan (if-else) penentuan tema

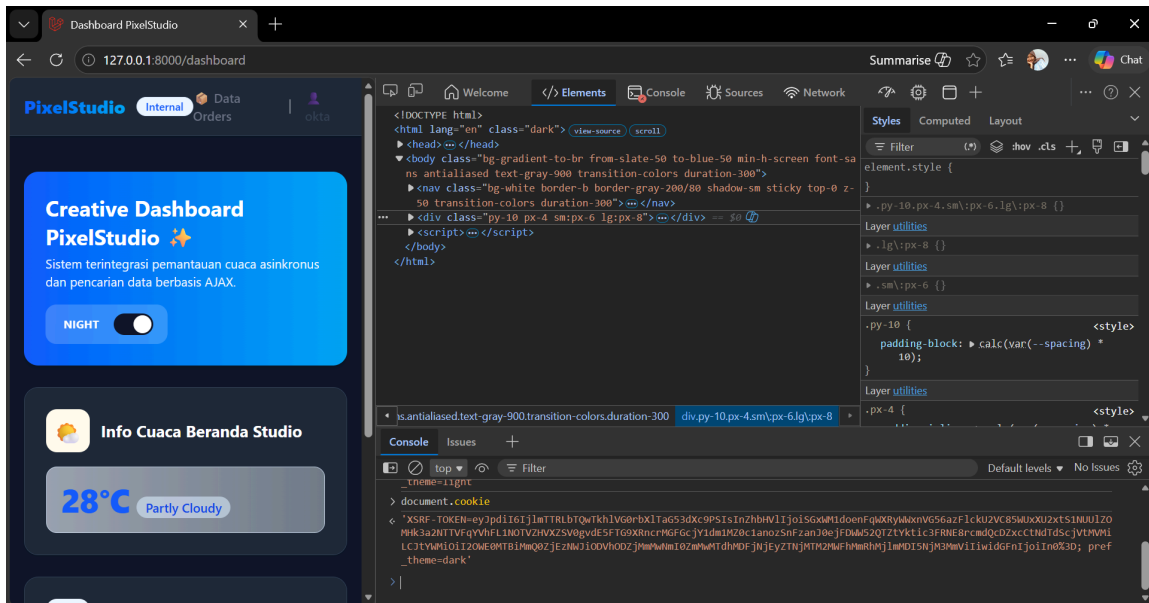
```

1  lang="en">
51  y class="bg-gradient-to-br from-slate-50 to-blue-50 min-h-screen font-sans antialiased text-gray-900 transition-colors duration-300"
138  <script>
139
140  const toggleInput = document.getElementById('theme-toggle');
141  const themeText = document.getElementById('theme-text');
142
143  function applyTheme(theme) {
144    document.cookie = `pref_theme=${theme}; max-age=${60*60*24*30}; path=/`;
145
146    if (theme === 'dark') {
147      document.documentElement.classList.add('dark');
148      themeText.innerText = "Night";
149      toggleInput.checked = true;
150    } else {
151      document.documentElement.classList.remove('dark');
152      themeText.innerText = "Light";
153      toggleInput.checked = false;
154    }
155  }
156
157  toggleInput.addEventListener('change', function() {
158    const targetTheme = this.checked ? 'dark' : 'light';
159    applyTheme(targetTheme);
160  });

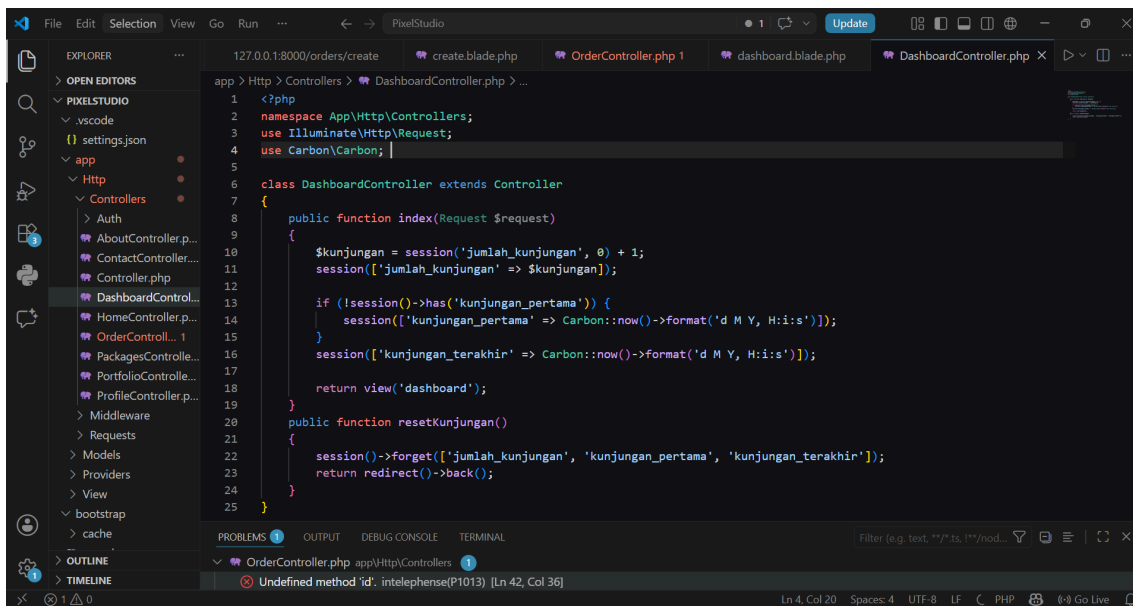
```

f. Menampilkan pesan sukses / perubahan visual





4. Penerapan State Management Menggunakan Session Server-Side (Laravel) untuk Manajemen Pelacakan Aktivitas Kunjungan Pengguna Berbasis Waktu Nyata.



```
2         ang="en">
51         lass="bg-gradient-to-br from-slate-50 to-blue-50 min-h-screen font-sans antialiased text-gray-900 transition-colors d
67         v class="py-10 px-4 sm:px-6 lg:px-8">
68         <div class="max-w-4xl mx-auto space-y-6">
135         <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
139             <span class="text-2xl">
140             </div>
141             <div>
142             <h3 class="text-lg font-bold text-gray-800">Statistik Kunjungan Sesi</h3>
143             </div>
144         </div>
145
146         <form action="/reset-kunjungan" method="POST">
147             @csrf
148             <button type="submit" class="text-xs bg-rose-50 text-rose-600 hover:bg-rose-100 font-bold px-3 py-2
149             Reset Hitungan
150             </button>
151         </form>
152     </div>
153
154     <div class="grid grid-cols-1 sm:grid-cols-3 gap-4">
155     <div class="bg-slate-50 p-4 rounded-xl border border-gray-100 text-center">
156     <span class="text-xs text-gray-400 block font-medium uppercase tracking-wider mb-1">Total Kunjungan< span>
157     <span class="text-2xl font-black text-emerald-600">{{ session('jumlah_kunjungan', 0) }} Kali</span>
158     </div>
159     <div class="bg-slate-50 p-4 rounded-xl border border-gray-100">
```

```
27     Route::middleware(['auth'])->group(function () {
50         Route::post('/api/save-preferences', function (Request $request) {
61             <div class="max-w-4xl mx-auto space-y-6">
62             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
63             <div class="max-w-4xl mx-auto space-y-6">
64             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
65             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
66             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
67             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
68             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
69             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
70             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
71             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
72             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
73             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
74             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
75             <div class="bg-white rounded-2xl p-6 shadow-sm border border-blue-100/60 transition-all duration-300">
```

